

THOMAS MICHAEL HENDERSON

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PROFESSIONAL SUMMARY

Research Engineer with 3 years of experience in machine learning and data systems development. I've spent my career building experimental prototypes and optimizing algorithmic solutions for real-world problems. I'm particularly interested in bridging the gap between academic research and production systems, and I enjoy working in cross-functional teams where I can contribute both technical depth and practical problem-solving skills.

WORK EXPERIENCE

Research Engineer

Nexus Intelligence Ltd

nexusintel.io

Birmingham, United Kingdom

March 2023 - Present

Designed and implemented a novel neural network architecture for time-series forecasting that reduced prediction error by 23% compared to baseline LSTM models.

Conducted experimental validation across 12 proprietary datasets, documenting findings in technical reports and contributing to 2 peer-reviewed publications.

Optimized data pipeline for model training, reducing preprocessing time from 8 hours to 45 minutes through vectorization and parallel processing.

Collaborated with product team to deploy research prototype into staging environment, ensuring code quality and reproducibility standards were maintained.

Machine Learning Engineer

DataCore Systems

datacore.co

Coventry, United Kingdom

August 2021 - February 2023

Developed feature engineering pipeline in Python that extracted 150+ meaningful variables from raw sensor data for predictive maintenance models.

Implemented experiment tracking system using MLflow, allowing the team to version 40+ models and compare performance metrics systematically.

Built data validation suite using pytest and pandas, catching 18 data quality issues in production that would have skewed model results.

Presented quarterly findings to stakeholders, translating complex technical results into actionable business insights for 8+ decision makers.

Junior Research Analyst

Applied Technologies Research Group

atresearch.net

Birmingham, United Kingdom

June 2020 - July 2021

Conducted literature review of 200+ academic papers on reinforcement learning applications, summarizing findings into structured knowledge base.

Prototyped 5 different algorithmic approaches for optimization problems, benchmarking each against standard baselines and documenting trade-offs.

Maintained internal research documentation and GitHub wiki pages to ensure team knowledge remained accessible and up-to-date.

Assisted senior researchers with data collection and annotation tasks, processing 15,000+ training samples for supervised learning experiments.

EDUCATION

Master of Science in Computer Science

University of Birmingham

Birmingham, United Kingdom

Graduated: September 2020

Dissertation: Deep Learning Approaches for Anomaly Detection in Temporal Data

Bachelor of Science in Computer Science

University of Birmingham

Birmingham, United Kingdom

Graduated: June 2019

First Class Honours

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, Bash

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Neural Networks, Time-Series Analysis

Data Engineering: Pandas, NumPy, Apache Spark, SQL optimization, ETL pipelines

Tools and Platforms: Linux, Docker, Git, Jupyter Notebooks, AWS (EC2, S3, SageMaker), Weights & Biases

Methodologies: A/B testing, Experimental design, Statistical analysis, Model validation, Hyperparameter tuning

PROFESSIONAL CERTIFICATIONS

AWS Certified Cloud Practitioner

Issued: March 2022

LANGUAGES

English: Native

French: Intermediate